

at scale. The then-secret gadget would allow users to instantly cast content from mobile devices or laptops to television screens, the way similar products did at the time. But unlike other products, Google needed it to be small, sleek and at an ultra-competitive price point.

Flex engineers designed an octagonal-shaped, printed circuit board (PCB)—an unprecedented form factor—and solved other significant technological challenges, too. The design of the circuit board took advantage of the gaps between circuits to disperse heat so it would not burn the circuitry. Flex developed an antenna with



Google Chromecast

a ribbon nestled between the layers of the circuit board that could sniff out a signal from behind a flatscreen. Together, Google and Flex designed a working prototype in just four weeks.

In an age of disruption, where

time-to-market makes the difference between pioneers and laggards, innovation must happen on multiple fronts. Collaboration between Google and Flex brought Chromecast design to life and got the device into consumers' hands at a staggering pace.

The resulting size and curves differentiate Chromecast from its competitors—and so does the price. Google began selling Chromecast streaming sticks for US\$ 35 in July 2013, offering a more accessible alternative to competitors' US\$ 100+ dollar products. Flex manufactured seven million of the devices during the first four months of production!

VOICE TECHNOLOGY:

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"MORE THAN FIFTY PER CENT OF SEARCHES WILL BE VOICE BASED BY 2020"

Voice technology, speech recognition and deep learning have changed the face of search as we know it. Artificial intelligence (AI), machine learning (ML) and Big Data technologies are sure to help leverage the penetration and adaptability of voice-based search. Sahil Chopra, founder and chief executive officer, iCubesWire, in a conversation with Deepshikha Shukla, shares how voice-based search can improve digital advertising and search engine optimisation

Q. How does voice search technology work?

A. Voice search works on the natural language processing (NLP) process, which simply defines how a computer comprehends human language, and AI. This technology has transformed how humans interact with technology.

Voice technology is the next-generation disruptor in business intelligence, marketing and advertising. All smartphones these days are equipped with voice search such that you can use Google without using the keyboard, for instance.

Q. What are the benefits of voice search?

A. There is certainly consumer comfort with voice-generated search. A user gets instant gratification from using voice, which allows him or



Sahil Chopra, founder and CEO, iCubesWire

her to carry out searches on the Web or shop on platforms like Amazon, without having to type a single letter.

Then of course, the experience is more conversational and interpersonal in the more conventional sense.

Voice search is definitely impacting the way advertising is seeing a transformational phase. Voice as search is enabling businesses to interact with its target groups, and it will play a crucial role in unifying branded content marketing made available by digital marketers across the globe.

Q. How can a voice search strategy improve digital advertising?

A. A voice search strategy will change the digital advertising landscape like never seen before. It is a watershed movement and voice search will only delve deeper into our searches in the coming years.

This is another crucial reason why companies invest heavily in such technology today, so that users do